Second Sprint: Literature Review

Due Oct 30 · Sprint review Monday, November 2

Marcus Birkenkrahe

COSC 3430 · Machine Learning · Augustana University

Fall 2026

What is a literature review?

What is a literature review?

- You find out **what others have already done** on your question before you build anything yourself: which approaches exist, which datasets and metrics people use, what results they got.

What is a literature review?

- You find out **what others have already done** on your question before you build anything yourself: which approaches exist, which datasets and metrics people use, what results they got.
- **It's not a list of links.** For each source you say what it does, how it does it, and what you take from it for your project.

What is a literature review?

- You find out **what others have already done** on your question before you build anything yourself: which approaches exist, which datasets and metrics people use, what results they got.
- **It's not a list of links.** For each source you say what it does, how it does it, and what you take from it for your project.
- **The synthesis is the part that counts:** what is known, which approach you choose and why, and what your project adds.

What is a literature review?

- You find out **what others have already done** on your question before you build anything yourself: which approaches exist, which datasets and metrics people use, what results they got.
- **It's not a list of links.** For each source you say what it does, how it does it, and what you take from it for your project.
- **The synthesis is the part that counts:** what is known, which approach you choose and why, and what your project adds.
- **If you use AI**, reconcile your hypothesis ("What may be useful for me?") with the AI's response ("This is useful for you!").

Why now?

Why now?

- Your research question gets **smaller and sharper** once you know what's been done. Many first-sprint questions were too broad: aim for a **minimum viable product (MVP)** first, stretch goals second.

Why now?

- Your research question gets **smaller and sharper** once you know what's been done. Many first-sprint questions were too broad: aim for a **minimum viable product (MVP)** first, stretch goals second.
- You learn **which metric and which baseline** to compare against. Without a baseline, you can't say whether your model is any good.

Lessons from the first sprint

Lessons from the first sprint

- **A link is not a reference:** complete every entry (author, title, year, source) and stay with one style (APA, MLA, Chicago, or Harvard).

Lessons from the first sprint

- **A link is not a reference:** complete every entry (author, title, year, source) and stay with one style (APA, MLA, Chicago, or Harvard).
- **Disclose your AI use, always** – even if it's "none used". Your statement and your references must agree.

Lessons from the first sprint

- **A link is not a reference:** complete every entry (author, title, year, source) and stay with one style (APA, MLA, Chicago, or Harvard).
- **Disclose your AI use, always** – even if it's "none used". Your statement and your references must agree.
- **AI-suggested sources that nobody checked:** you must have read every source you cite. A source that does not exist costs points.

What counts as a source?

What counts as a source?

- **At least five sources, at least two authoritative:** a book, a peer-reviewed article, official documentation, or the documentation of your dataset or library.

What counts as a source?

- **At least five sources, at least two authoritative:** a book, a peer-reviewed article, official documentation, or the documentation of your dataset or library.
- **Your dataset is a source too:** who published it, when, where.

What counts as a source?

- **At least five sources, at least two authoritative:** a book, a peer-reviewed article, official documentation, or the documentation of your dataset or library.
- **Your dataset is a source too:** who published it, when, where.
- **Blogs and videos are fine**, but not as your only sources. AI conversations do not count toward the five.

Where to look

Where to look

- **Google Scholar, arXiv**, the dataset's own page (Kaggle, UCI), the documentation of the libraries you use, and the textbook (Géron).

Where to look

- **Google Scholar, arXiv**, the dataset's own page (Kaggle, UCI), the documentation of the libraries you use, and the textbook (Géron).
- **Find a literature review** (a survey) on your topic: someone has already collected and compared the important sources.

Where to look

- **Google Scholar, arXiv**, the dataset's own page (Kaggle, UCI), the documentation of the libraries you use, and the textbook (Géron).
- **Find a literature review** (a survey) on your topic: someone has already collected and compared the important sources.
- **Litmaps** (`litmaps.com`): start from one paper and see a map of the papers it cites and that cite it.

Where to look

- **Google Scholar, arXiv**, the dataset's own page (Kaggle, UCI), the documentation of the libraries you use, and the textbook (Géron).
- **Find a literature review** (a survey) on your topic: someone has already collected and compared the important sources.
- **Litmaps** (`litmaps.com`): start from one paper and see a map of the papers it cites and that cite it.
- **OpenResearch** (`github.com/alphaXiv/OpenResearch`): an open-source workspace for AI research agents that review the literature.

Where to look

- **Google Scholar, arXiv**, the dataset's own page (Kaggle, UCI), the documentation of the libraries you use, and the textbook (Géron).
- **Find a literature review** (a survey) on your topic: someone has already collected and compared the important sources.
- **Litmaps** (`litmaps.com`): start from one paper and see a map of the papers it cites and that cite it.
- **OpenResearch** (`github.com/alphaXiv/OpenResearch`): an open-source workspace for AI research agents that review the literature.
- **Use AI to find candidates and to summarize**, then open and read the source yourself, and verify the citation.

Sprint review

Monday, November 2

in class

Sprint review

Monday, November 2

in class

- Every team member presents **at least one** of the sources and answers questions about it.

Sprint review

Monday, November 2

in class

- Every team member presents **at least one** of the sources and answers questions about it.
- **Questions?** Ask me early, not at the end of the sprint!